

Traffic Behavior Profiling & Intelligence Report

Analysis Mode: Full

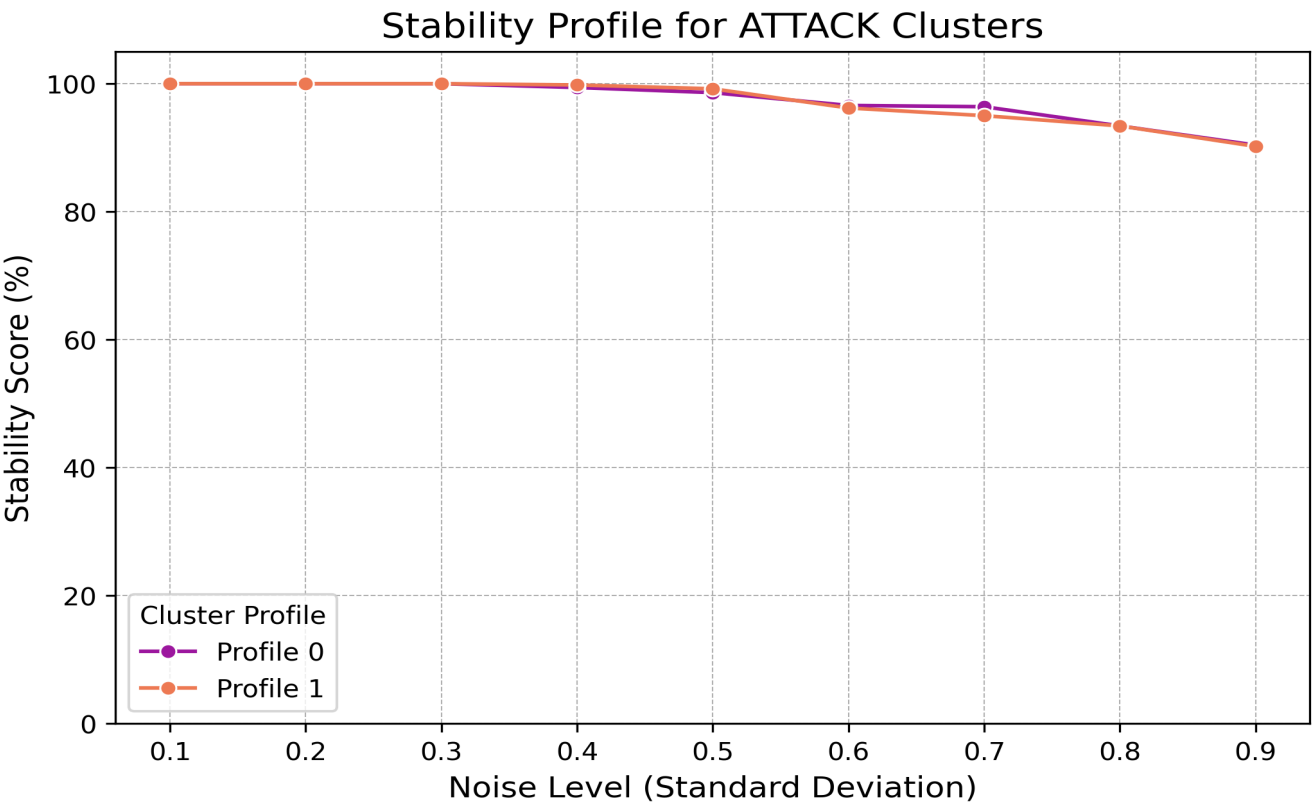
Executive Summary

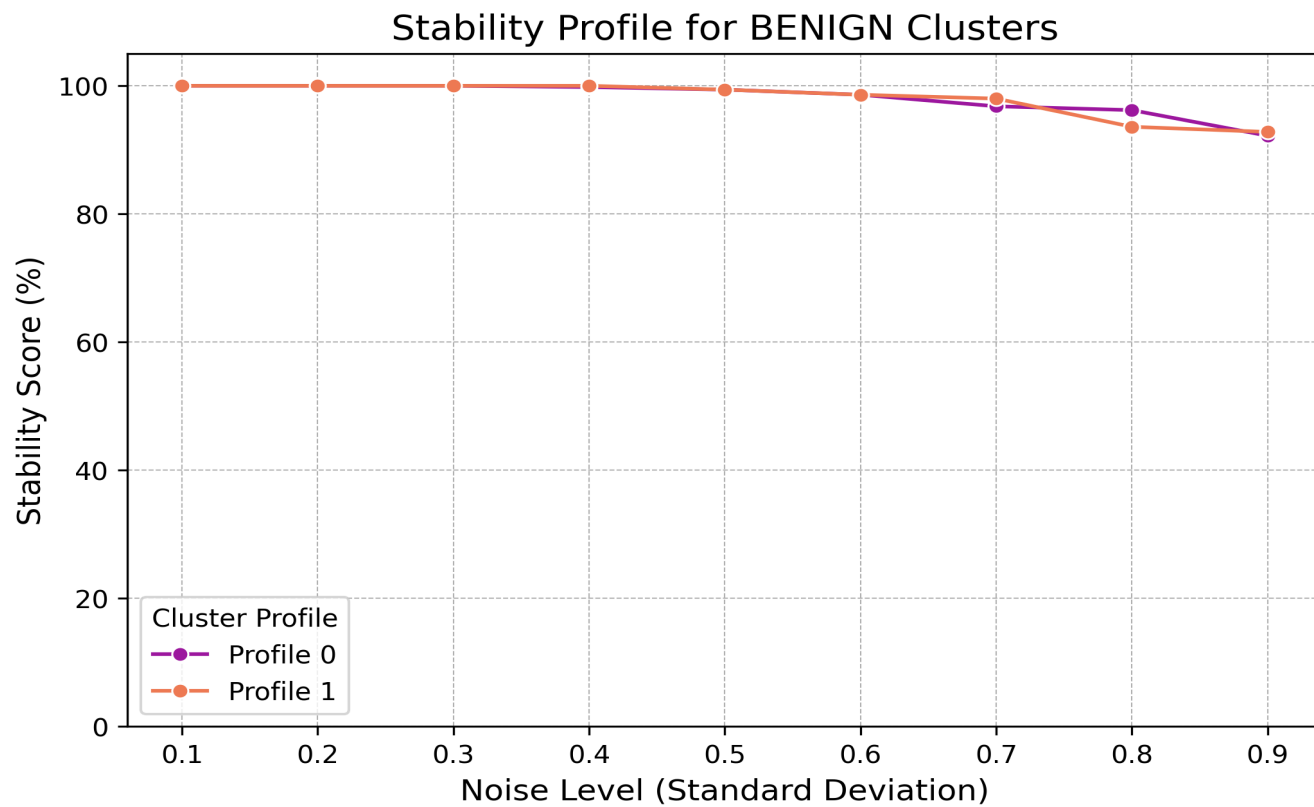
This report details a scalable, multi-stage data analysis pipeline designed to transform raw network data into **actionable cybersecurity intelligence**. Leveraging a distributed computing framework (Apache Spark), the system processed 300000 network records to automatically discover and profile distinct malicious and benign activities. The analysis pipeline consisted of: **(1) Data Ingestion and Preprocessing**, **(2) Feature Selection** using a tuned Decision Tree, **(3) Behavior Profiling** using a tuned K-Means clustering algorithm, **(4) a tuned Mitigation Recommendation Engine**, and **(5) Profile Stability Validation** via a Monte Carlo stress test. The key features for distinguishing traffic included **flow_duration, destination_port, total_fwd_packets**. This project demonstrates a practical approach for Security Operations Centers (SOCs) to automate threat analysis at scale.

Automated Mitigation Recommendations

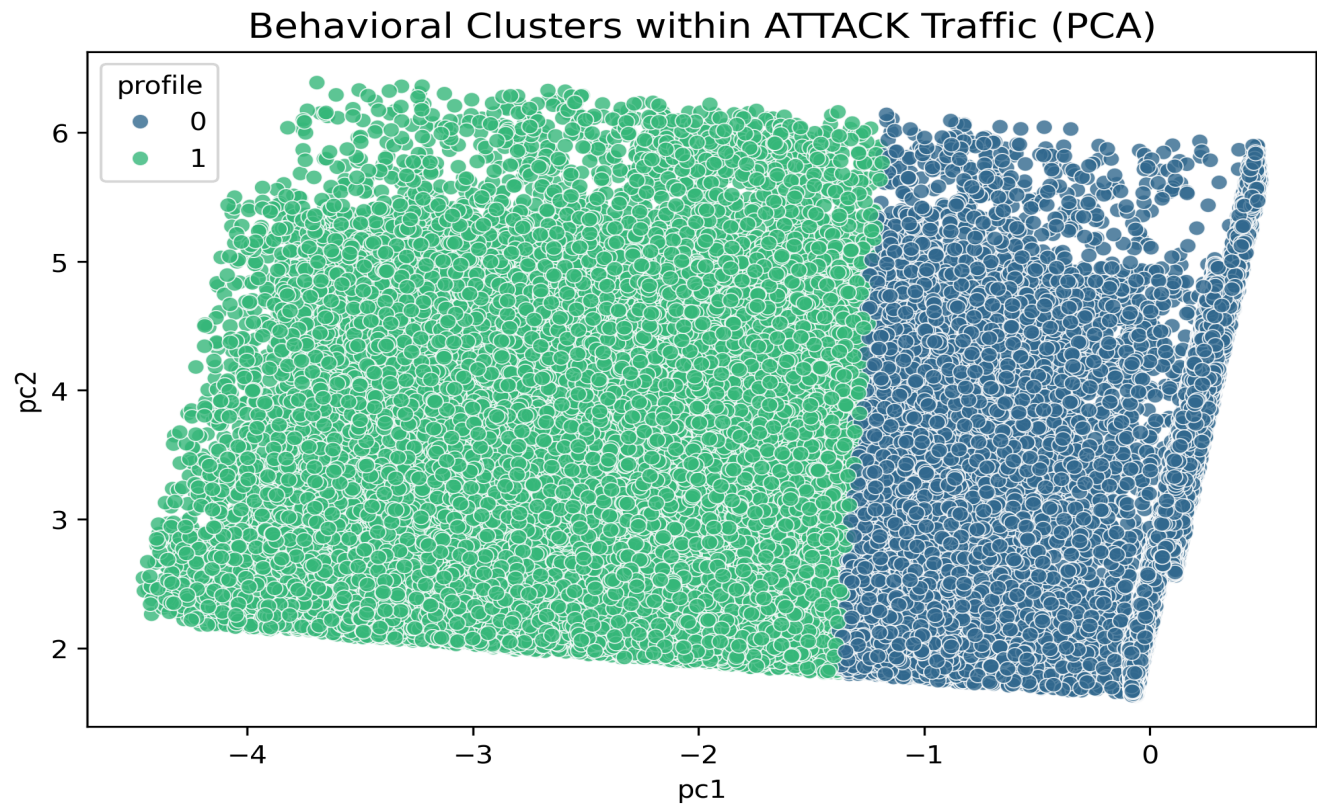
Threat Profile	Recommended Action 1	Recommended Action 2	Recommended Action 3
ATTACK_P0	Isolate-Host	Blacklist-IP	Analyze-Payload
ATTACK_P1	Blacklist-IP	Isolate-Host	Analyze-Payload

Profile Stability Stress Test (Monte Carlo)





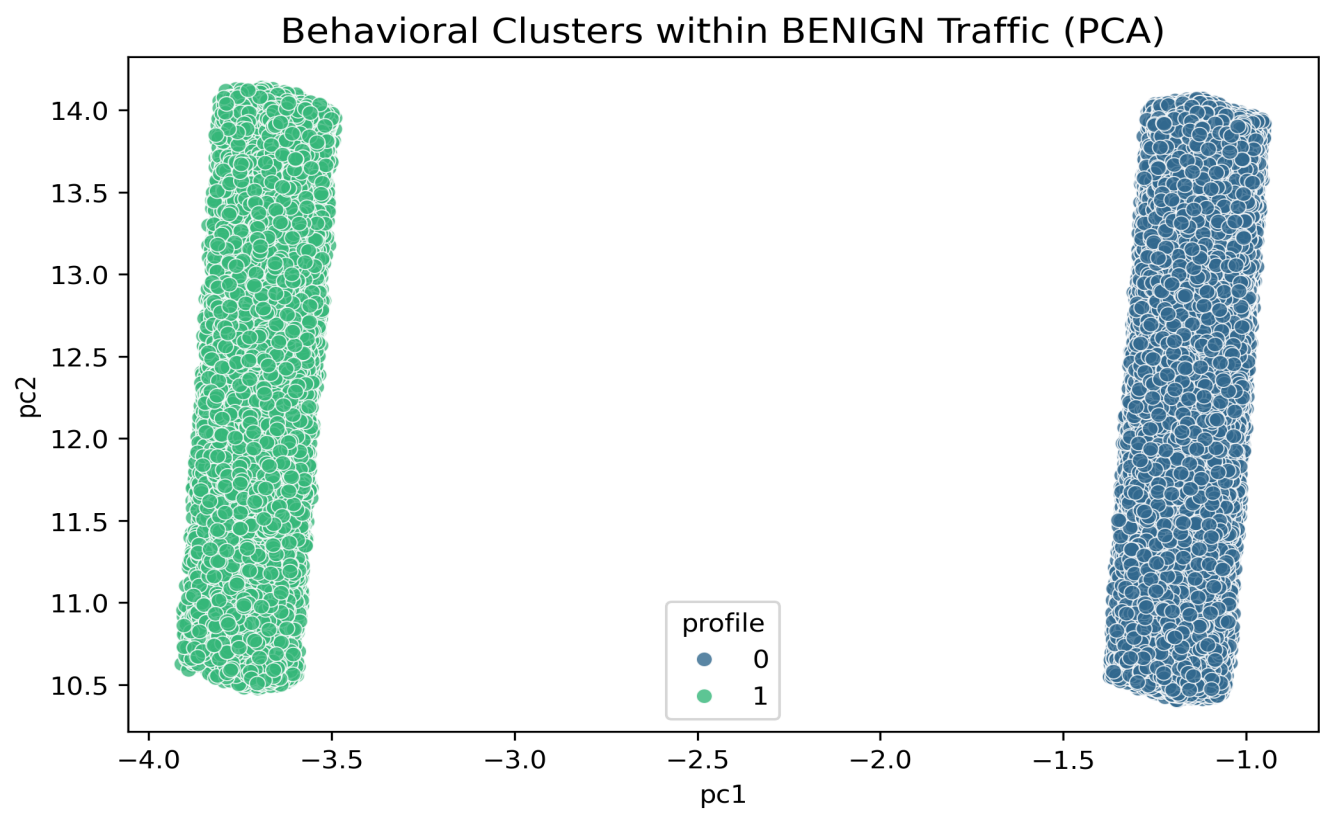
Detailed Profile: ATTACK



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1
Flow Duration	336,975.48	2,890,267.69
Destination Port	17,143.29	14,685.10
Total Fwd Packets	2.09	11.38

Detailed Profile: BENIGN



Profile Fingerprints (Cluster Centers)

Feature	Profile 0	Profile 1
Flow Duration	2,443,822.67	2,465,895.59
Destination Port	22,368.45	22,554.19
Total Fwd Packets	9.81	9.77